Evaluating Local Employment and Worker Commute Patterns Across California State Prisons

Jade Jaimes Chavez
Kampe Rushoka

Research Question

This exploratory data analysis investigates the spatial extent of California prison employees' daily commutes in 2023 to evaluate whether state prisons draw their workforce from immediate local communities.

Background

This research is informed by the claim that prisons contribute significantly to its surrounding local communities' economy.[1] The Urban Institute, named for example Blythe, California's local community which demanded the prison to delay their shut-down to allow employees time to find other jobs..[2] On the other hand, opposing research claims prisons draw their workforce from outside the local community and rather impeded economic growth[3] Thus, analyzing prison employee demographics and commute patterns in California can help guide policies that foster real local economic growth.

Data Inventory

To answer our research question, we used the Homeland Infrastructure Foundation-Level Data (HIFLD) Prison Boundaries, a geospatial dataset containing secure detention facilities in the United States and some U.S. territories, Our second data source was the United States Census Bureau, **TIGER/Line Geodatabases**, which provides California geodatabase for 2020 to get census block level data Lastly, we used the Census Block **Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES)**, which provides information on each home-workblock data, representing where people live and work at a census block level.

How the Data was Handled

Our data was prepared in four stages. First, the HIFLD dataset was filtered to California facilities of "STATE", and kept only those containing "State Prison" in their name or matching a list of CDCR institutions. Names were standardized, and their establishment years were added by looking it up. Second, we mapped 2020 California Census block polygons to each prison's coordinate system, spatially joining them so that every block intersecting a prison polygon was assigned to that prison. Third, we loaded the LODES OD data for 2022 to include all jobs using the code JT00. We also combined the "main" file and the "aux" file, with geocodes zero-padded to 15 digits. We loaded the WAC to validate the totals against the OD file, and to support the block-assignments we fixed. Fourth, we corrected the prison-to-block mapping by:

---

[1] https://www.prisonpolicy.org/scans/building.html
[2] https://www.urban.org/urban-wire/curbing-rural-prison-demand-and-responsibly-closing-prisons
[3] https://ideas.repec.org/a/bla/socsci/v85y2004i1p37-57.html

- We had five prisons where the HIFLD polygon missed the block holding the jobs. For those, we added nearby blocks that were within 1 km and were at least 80% public administration jobs. For example, the California Medical Facility had a block that was mostly health-care, and so it was added manually.
- Folsom and New Folsom were merged into one site because they share one block
- We flagged sites where public administration jobs made up less than 70% of block jobs, and called them mixed sites.

OD records were joined to the corrected mapping to create a worker-level table, where each row is the number of jobs from one home block to one prison. We used (euclidean) straight-line distances (in miles) between home and work block centroids in the California Albers projection (EPSG:3310). Out-of-state workers don't have distances, so we excluded them from averages. Finally, we aggregated the data by prison into weighted mean and median distances, shares within 10 miles, shares within 50 miles, and within the same block. These were all done along with age, earnings, and industry job counts.

Exploratory Data Analysis

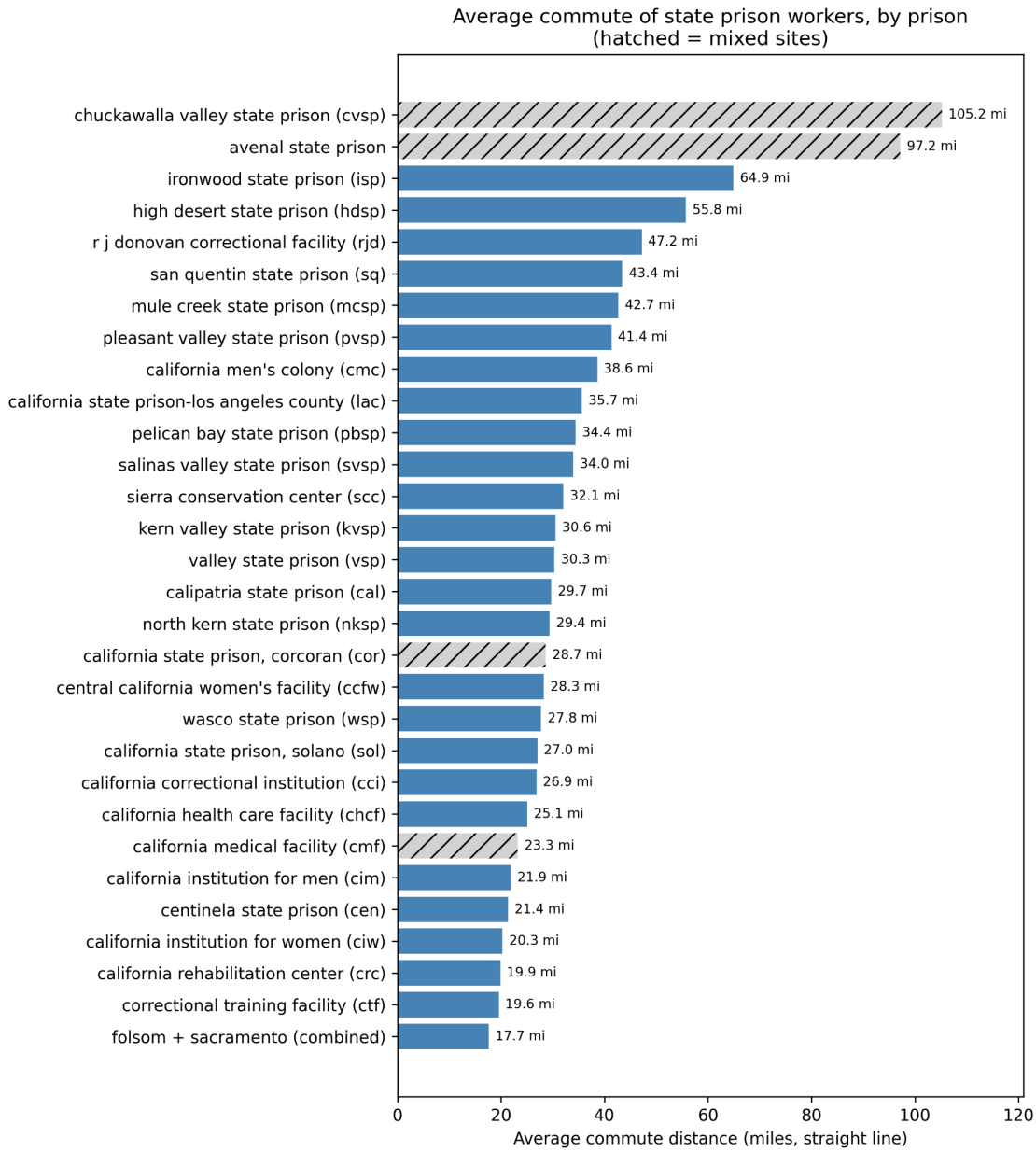
I. Commuting patterns of California Prison workers:

Statistic	Number of prison sites (N)	Mean	Standard deviation	Min	25th percentile (Q1)	Median (Q2)	75th percentile (Q3)	Max
Weighted Average Commute Distance (miles)	30	36.67	20.61	17.66	25.52	30.02	40.67	105.21

Table 1:Descriptive statistics of California Prisons Weighted Average Commute Distance

Across the 30 analyzed prison sites, the mean weighted average commute distance is 36.67 miles, while the median facility-level average is 30.02 miles. Regional transportation research on commuting in Southern California treats trips exceeding 20 miles as long-distance commutes due to their association with elevated commuting stress[4]. Using 20 miles as a benchmark for local employment, both the mean and median distances across California prisons lie well above this threshold. While there is substantial cross-site variation, indicated by a standard deviation of 20.61 miles and an interquartile range (IQR) of 15.15 miles, the middle 50% of prison average commutes still fall between 25.52 and 40.67 miles. Because even this lower quartile bound ($Q_1 = 25.52$ miles) exceeds 20 miles, these distribution metrics consistently indicate that average California prison worker commutes are non-local.

[4]https://its.uci.edu/research_products/working-paper-commuting-stress-ridesharing-and-gender-analyses-from-the-1993-state-of-the-commute-study-in-southern-california/

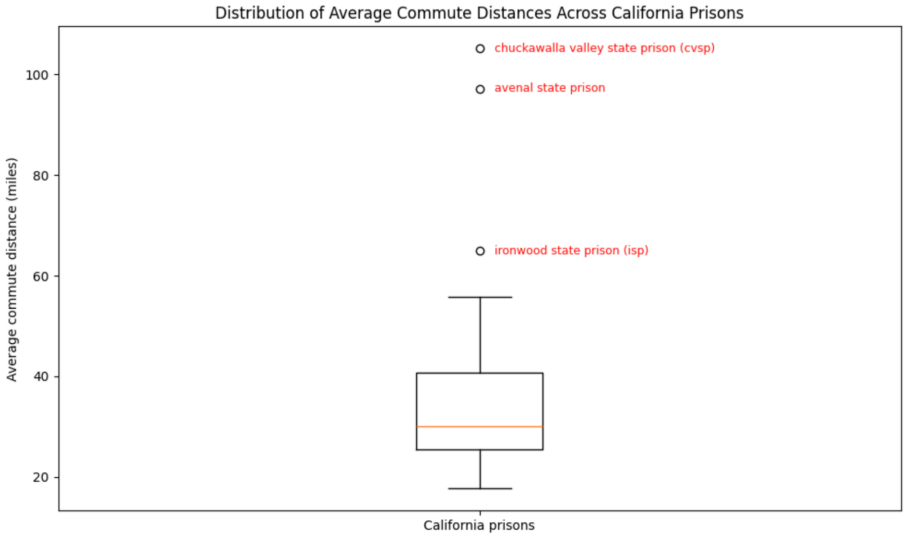


Graph 1: Weighted mean straight-line distance between workers' residential and workplace Census block centroids, and hatched bars indicate mixed-employment sites.

Graph 1 displays weighted average commuting distances across California state prison sites. Facilities with the highest commutes include Chuckawalla Valley (105.2 mi), Ironwood (97.2 mi), Avenal (64.9 mi), and High Desert (55.8 mi). These straight-line distances translate to roughly 45 minutes to two hours of driving, excluding traffic. Because centroid estimates measure straight-line paths rather than road networks, workers' actual travel routes are likely even longer.

Hatched bars identify mixed-employment sites where public administration accounts for under 70% of block-level jobs, meaning employment data cannot be attributed entirely to the prison. Because top-distance facilities like Chuckawalla Valley and Avenal are mixed sites, their estimates require cautious interpretation, as non-prison employers sharing those Census blocks may inflate average commuting distances.

Potential prison outliers/anomalies

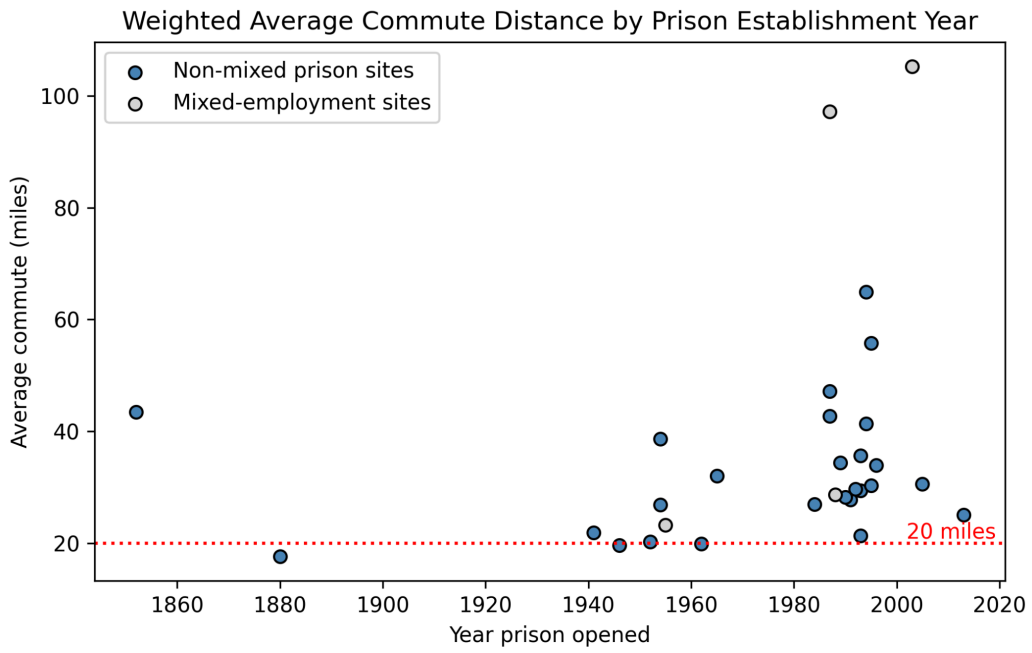


Graph 2: Box plot of Distribution of Average Worker Commutes Across California Prisons with Top Outliers

Graph 2 shows three outlier prisons, Chuckawalla Valley State Prison, Avenal State prison and Ironwood State Prison(which are 1.5 times above the IQR) with significantly high commuting distance. Interestingly for all three outliers, less than 2.2% of their workforces commute 10 miles or less, while 56.9% - 64.8% commute above 50 miles. Interestingly, the outlier of Avenal State Prison the only facility with 670 jobs recorded in trade, transportation, and utilities industry, with all other facilities having zero to none in that category virtually absent at other sites. The high industry concentration is likely due to Avenal's programs of free transportation for released inmates with Greyhound and Transmetro services[5]. A large share of Avenal's recorded workforce likely consists of these program drivers, workers who typically live far away and cover long distances for their jobs, which explains why Avenal stands out as a high-commute anomaly.

III. Commuting Distance and Establishment year of California prison:

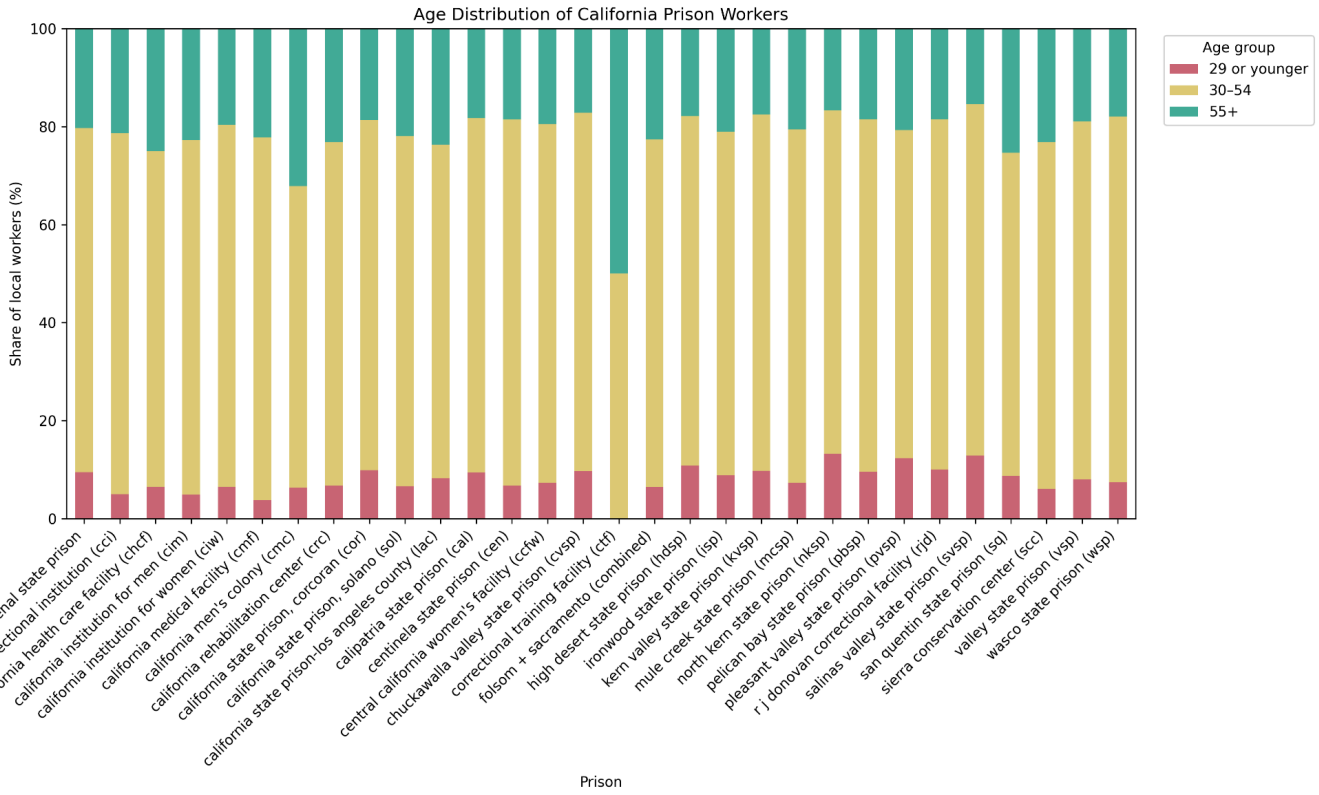
[5]https://sanquentinnews.com/transmetro-offers-free-transportation-to-prisoner-families/#:~:text=Southern%20region%20pickups%20include%20San%20Diego%2C%20Long,and%20water%20will%20be%20provided%20by%20TransMetro.



Graph 3: Scatterplot of Weighted Average Commute Distance in miles by Prison Establishment Year

Graph 3 shows that older prisons established before the 1970s were all near the 20-mile local commute threshold, whereas almost all facilities built between the 1980s and 2000s lie well above 20 miles. This disparity may stem from the state's prison expansion boom in the 1980s and 1990s, during which facilities were increasingly constructed in remote, rural areas that probably required workers to commute to farther rural region However, when we calculated the correlation, establishment year exhibited a very weak linear relationship with average commuting distance among non-mixed sites ($r = 0.19$).

IV. California Prisons Workforce Age.



Graph 5: 100% Stacked Bar Chart of Relative Age Distributions Among Local Workers Across California State Prisons

Graph 5 shows that the vast majority of California prison workers are aged 30–54, while young adults (29 or younger) represent a very small proportion across all. This might due to the fact that Younger Californians may be deterred by the long commutes, of which the average commute is nearly 37 miles, though **we lack evidence in our study for such deterrence.** Nonetheless, these combined findings strongly suggest that California prisons likely do not employ both local residents and younger demographics.

Conclusion and Limitations

Our analysis finds that California state prison sites have a high average weighted commute distance of 36.67 miles, which is significantly above the 20 miles threshold from existing research. Suggesting that California prison-related employment does not necessarily benefit nearby residents. Although newer prisons appeared to have longer commutes, we found no statistically significant correlation between prison establishment year and commute distance. Additionally we found that the workforce was predominantly aged 30–54, and very little workforce aged between 18-29, raising questions about whether younger workers face barriers to accessing these jobs due to commuting distance, although our analysis cannot establish this relationship.

Several limitations qualify these findings. First, mixed-sites most likely include jobs unrelated to prisons, which means we cannot make confident conclusions on them. Second, Folsom and New Falsom were combined as their locations could not be distinguished at the block level. Third, Census block centroids provide only approximate straight-line distances, not actual road distances. Finally, our prison-level analysis cannot establish causal relationships or directly measure local economic benefits.

Our findings suggest that policymakers should be cautious about justifying new prison construction as a local economic development strategy. Creating jobs does not necessarily mean those jobs are accessible to nearby residents. Policymakers should evaluate who fills these positions and whether employment benefits reach surrounding communities before treating prison construction as a means of stimulating local economies.

Github repository:
https://github.com/irushoka-lang/ca-prison-commute-analysis/tree/main